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SP25-BSE-082
Cal #1 ; Quiz #1

1) Solve

$$\frac{x}{2} + 9 < \frac{x}{4} + \frac{11}{2}$$

Sol.

$$9 - \frac{11}{2} < \frac{x}{4} - \frac{x}{2}$$

$$\frac{18-11}{2} < \frac{x-2x}{4}$$

$$\frac{7}{\cancel{2}_1} \times \cancel{4}^2 < x - 2x$$

$$14 < -x$$

$$\boxed{-14 < x}$$ sol.

2) Find domain set:

$$f(x) = \frac{2}{\sqrt{4x-5}}$$

$$= \frac{2}{\sqrt{4(0)-5}}$$

$$= \frac{2}{\sqrt{-5}}$$

$\boxed{\text{Domain set} = \{R\} \cancel{\{R, 0\}}}$ $\{R \geq 2\}$

$\{R\} - \frac{5}{4}$

3) Solve

$$\lim_{x \to 1} \frac{x^2+x-2}{2x^2-2x}$$

$$= \frac{x^2+x-2}{2x(x-1)}$$

putting limits: $\frac{(1)^2+1-2}{2(1)(1-1)} = \frac{2-2}{2(0)} = \frac{0}{0} \neq$

) $\lim_{x \to 0} \frac{\sin(\sqrt{3}\,x)}{5x}$

Putting limits:

$$\frac{\sin\sqrt{3}(0)}{5(0)} = \frac{\sin(0)}{0} = 0 \text{ ans.}$$